

Mini Project Report: 2025-26 Session

ATM Management System

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PROJECT FOUNDATIONS

Acknowledgement

Heartfelt gratitude to Prof. (Dr.) Sonal Sharma and Prof. (Dr.) Sameer Dev Sharma for fostering a healthy environment. Special thanks to mentor **Mr. Sanjay Mishra** for invaluable technical guidance.

Declaration

I, Nikhil Rawat, hereby declare that the "ATM Management System" is an original work conducted under the supervision of USCS Department, Uttarakhand University.

THE BANKING EVOLUTION

Introduction

In a fast-paced digital economy, Automated Teller Machines (ATMs) are vital for 24/7 banking. This project simulates a secure environment for account management using the C language.

The Need

- Eliminating long bank queues.
- Providing instant cash accessibility.
- Reducing manual human errors.
- Ensuring real-time balance tracking.



CORE OBJECTIVES



Secure Access

Implementing robust PIN-based authentication and lockout mechanisms.



Admin Control

Providing backend tools for account creation and real-time monitoring.



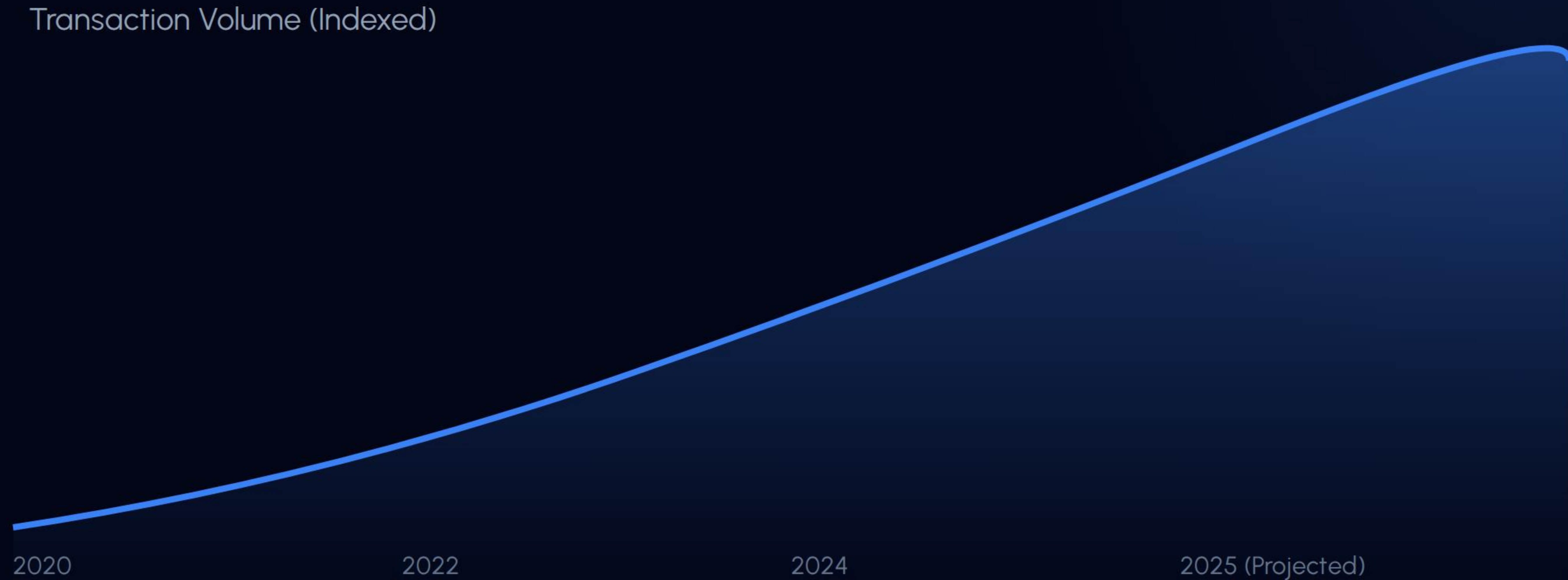
Operational Efficiency

Speedy transaction processing for withdrawals, deposits, and transfers.

DEVELOPMENT ROADMAP



RISE OF DIGITAL BANKING



Automated banking services are seeing an exponential rise in user adoption globally.

SYSTEM ARCHITECTURE: DFD

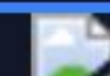
Data Flow Diagram Architecture

Data Flow Logic

The system utilizes a structured hierarchy to process information between the User, the ATM Core, and the Bank Server.

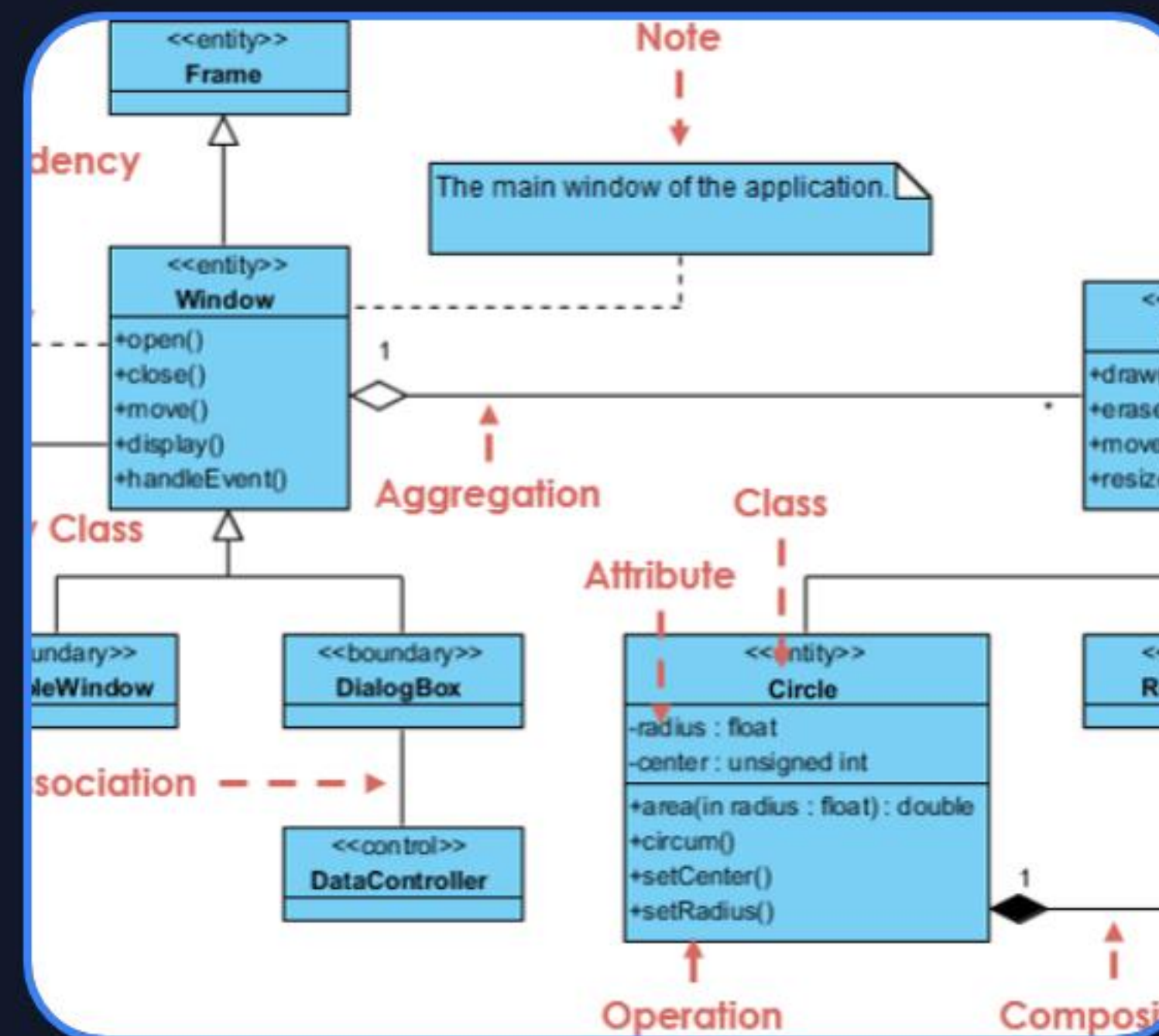
- **Inputs:** PIN, Transaction Type, Amount.
- **Processing:** Authentication, Balance Verification.
- **Outputs:** Cash, Receipt, Transaction Log.

MODELING LOGIC

 Activity Diagram

Activity

Flow from card insertion to receipt.



Class

Structural data relationships in C.

 State Diagram

State

Lifecycle from Idle to Authenticated.

TECHNICAL SPECIFICATIONS

Requirement Type	Specification Detail	Purpose
Language	C Programming	High-performance logic execution
Environment	Windows 10 / Dev C++	Standardized IDE development
Database	Binary Files (.dat)	Persistent storage for user records
Hardware	Core i3 / 2GB RAM	Minimum operational footprint

SECURITY & RELIABILITY

3

Max PIN Attempts

Protocol Highlights

Security is the foundation of our banking system. We have implemented critical measures to ensure user trust and data integrity.

- **Account Lockout:** Automatic suspension after 3 fails.
- **Audit Trail:** Every transaction is logged with time/date.
- **Secure Storage:** PINs are never displayed in plain text.

LOOKING AHEAD



Biometrics

Integrating fingerprint & facial recognition to replace traditional PINs.



App Sync

Mobile-based OTP authentication for two-factor security.



AI Fraud

Predictive algorithms to detect unusual spending patterns.



Thank You

Any Questions?

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IMAGE SOURCES



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